

GERARDO ANTONIO CORRAL RUIZ

AI/ML Applied Research & Advisory • Data Science • Master's candidate in Digital Innovation & Governance

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PROFESSIONAL SUMMARY

Master's candidate (**LM-91: Methods and Techniques for the Information Society**) at the **University of Bologna**, with a focus on **machine learning and artificial intelligence with a critical approach to governance, cyber risk management, and digital transformation**. I have developed **machine learning projects for industrial cybersecurity, particularly in IIoT/OT contexts**. I have also carried out university projects in **public health based on open data from sources such as ISTAT, Open-Meteo, and Kaggle**. I continually develop my **data science skills using Python and R**. Proficient in **Agile methodologies (Scrum)**, with experience using tools such as **Notion, Taiga, Essence Workbench, Slack, and Airtable**. As a **university lecturer**, I lead **contemporary philosophical debates with my students on technology and digital ethics**, linking these topics to the **interactions between technology, society, and governance**.

PROFESSIONAL EXPERIENCE

AI/ML Applied Research & Advisory @ Syntezia Sàrl (Switzerland) 2026–Present

In my internship at Syntezia, I am conducting applied research on **Artificial Intelligence and Machine Learning for cybersecurity in IT and OT/IIoT environments**; particularly am developing **Intrusion Detection System approaches based on Machine Learning and Deep Learning, using industrial telemetry and network traffic**.

In addition, I analyze where, when, and under which constraints the solution creates value across the **Protect/Detect/Respond functions of the NIST CSF 2.0**, and I am carrying out an **ecosystem mapping activity involving EDIHs and potential partners**.

IFAB 4 Next Generation Talents @ Hypermeteo (Italy) 2026–Present

I have been selected to collaborate on the development of an **Artificial Intelligence assistant for Hypermeteo**, focused on the **analysis of meteorological data in NetCDF format, the detection of patterns and anomalies**, and the generation of actionable insights to **support the interpretation of complex climate information and decision-making**.

Academic Researcher (Mexico)

- **Researcher in Training**, "Racionalidad, acción y vitalidad de las instituciones sociales" 2025–present
- **Academic & Research Assistant** – Dr. Héctor Zagal Universidad Panamericana 2018–2023

University and High School Professor (Mexico)

- **Universidad Panamericana** – Course on **Selected Topics in Contemporary Culture** (online lectures) 2025–2026
- **Northridge School Mexico** – Full-time **Spanish and Philosophy** courses. 2023–2024

Product Owner @ PCI Lab (Mexico) 2020–2022

Development of a **serious game** with the Narrative Design Team to **promote emotional intelligence among 15–18-year-olds**. We adopted **Agile (Scrum) with weekly sprints**, using **Airtable** for the backlog, roadmap, and **Slack** for **real-time coordination**. The project received the **ILAN Innovation Award**.

EDUCATION

M.S. in Digital Innovation, Policies and Governance (LM-91) @ University of Bologna (Italy) 2024–2026

I was **elected class representative** in my Master's program. To date, my **GPA is 29.43/30**. Below are a few key highlights from the **main projects I've developed**.

- **Machine Learning for Cardiovascular Disease Detection (Laboratorio di Programmazione, INF-05/A, 2024)** **FINAL GRADE: 30/30**
 - End-to-end Data Science project on the Heart Disease Dataset (Kaggle).
- **Moderate and Extreme Temperatures and Climate–Health Risk in Bologna: A Data-Driven Study of Climate Risk (Metodi Informatici per la Trasformazione Digitale, INF-05/A, 2025)** **FINAL GRADE: 30/30 with Honours**
 - I identified high-risk days for vulnerable groups, providing targeted support for policy decision-making.

- **The Digital Panopticon of Artificial Intelligence: between Safety and Freedom (Governance e Diritto delle PPAA, IUS/09, 2024)** **FINAL GRADE: 30/30 with Honours**
 - I analyzed the impact of AI on public safety in Europe, with a particular focus on the EU AI Act, the GDPR, and Regulation (EU) 2024/1689.
- **Cyber Threats in IIoT: Can ML Mitigate Vulnerabilities and Prevent Harm? (Governance della Cybersecurity, ING-INF/05, 2025)** **FINAL GRADE: 30/30 with Honours**
 - I wrote a paper on cyber risk and governance in IIoT/OT environments and developed a Python case study on ToN-IIoT (Modbus telemetry), testing ML models (Decision Tree, Random Forest, SVM) for binary and multi-class attack classification, with hyperparameter tuning and evaluation (F1, FPR/FNR). The goal is to help SOC analysts reduce alert fatigue and enable managers to identify where security needs to be strengthened.

B.A. in Philosophy @ Universidad Panamericana (Mexico) 2018-2023

- Selected for an **academic exchange program** at the **University of Oviedo, Spain** (2022-2023)
- **President of the Student Council**, Faculty of Philosophy (2021 - 2022)

B.Eng. in Digital Animation Engineering and Video Game Programming (not completed) @ UP, Mexico 2017-2019

CONFERENCES & PUBLICATIONS

- Conference Speaker:** "Riesgos y amenazas emergentes de la digitalización y la IA en hospitales y servicios de salud" for the Mexican Red Cross - DIPEC **2026**
https://www.facebook.com/dipeccruzroja?locale=es_LA
- Conference Speaker:** "Identidad líquida en tiempos de redes sociales" - Instituto Antonio Nariño, **Colombia** **2024**
<https://www.filosofoteca.app/volunteer-speakers>
- Published an article on **generative AI and the future of the arts** in Generación Universitaria (**El Universal**). **2022**
<https://www.generacionuniversitaria.com.mx/tu-voz/dibujo-asistido-por-inteligencia-artificial-es-arte/>
- Moderator and Discussant**, XXIV Student Colloquium, School of Philosophy, Universidad Panamericana **2020-2021**

CERTIFICATIONS, COURSES & HACKATHONS

AI, DATA SCIENCE & BUSINESS ANALYTICS

- AI Automation Hackathon powered by n8n** - Jenia **2026**
- Intelligenza artificiale e business intelligence: strumenti e algoretica** - IFOA **2025**
- EITC/AI/MLP Machine Learning with Python** - European Information Technologies Certification Institute (EITCI) **2025**
- Seminario LEEIA Lógica, Epistemología y Ética de la Inteligencia Artificial** - Universidad Panamericana **2025**
- Strumenti di data analysis e visualization** - IFOA **2025**
- EITC/CP/PPF Python Programming Fundamentals** - European Information Technologies Certification Institute (EITCI) **2025**
- Business Analytics with Excel: Elementary to Advanced** - Johns Hopkins University **2023**
- What is Data Science?** - IBM **2023**
- PredictionX: Omens, Oracles & Prophecies** - HarvardX **2020**

MANAGEMENT & INNOVATION

- Management: A Professional Future** - IESE, University of Navarra, Barcelona **2023**
- Introducción a la Dirección de Empresas (IDE)** - IPADE Business School **2021**

DIGITAL ARTS, STORYTELLING & PUBLIC SPEAKING

- Awarded **2nd Place** in the "About AI" Category and **3rd Place** in the "Made by AI" Category - ALMA-AI Video Contest **2025**
- Story and Narrative Development for Video Games & Introduction to Game Design** - California Institute of the Arts **2022**
- Modern Arts & Ideas** - MoMA Certificate **2021**
- Studied Cinematography and Digital Post-Production** - SAE Institute **2019-2020**
- Theatre UP** - Universidad Panamericana **2017-2020**

HARD SKILLS

- **Agile Work:** Essence Workbench, Taiga, Notion, Slack, Airtable, Protégé, LaTeX
- **Data Analysis Tools:** Python (Pandas, Seaborn, Numpy, Sklearn, DoWhy), R, SQL, Git, JavaScript
- **Generative AI Tools:** ChatGPT, Gemini, NotebookLM, Sora, Midjourney, ComfyUI
- **Design and Video Editing Tools:** Adobe (Premiere, Photoshop, After Effects), Canva, Blender
- **Microsoft Office:** Word, PowerPoint, Excel

LANGUAGES

- **Native:** Spanish
- **Advanced:** English, Italian
- **Basic:** Japanese

I HEREBY AUTHORIZE THE PROCESSING OF THE PERSONAL DATA CONTAINED IN MY CV IN ACCORDANCE WITH ITALIAN LEGISLATIVE DECREE 196/2003 AND EU REGULATION 2016/679.